SUMIN LEE

M.S. Student, Mechanical Systems Engineering
Sookmyung Women's University, Seoul, Republic of Korea

Curriculum Vitae
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Research Interests

- **Robot Manipulation**: long-horizon and contact-rich manipulation
- **LLMs for Robotics**: LLM-based robot code generation and control
- **Robot Learning**: hybrid code–policy skill learning with reinforcement learning
- **Sim-to-Real Transfer**: real-to-sim verification of robot programs
- **Robot Perception**: multi-view perception and camera calibration

Education

2025.09 – Present **Sookmyung Women's University**, Seoul
M.S. in Mechanical Systems Engineering
GPA: 4.50/4.50 (Two semesters completed)
Advisor: Prof. Joo Yong Sim

2023.03 – 2025.08 **Sookmyung Women's University**, Seoul
B.S. in IT Engineering
GPA: 4.01/4.50

Publications

- **When to Code, When to Learn: Real-to-Sim-Verified Hybrid Code-Policy Skill Generation for Long-Horizon Robot Manipulation**
  Sumin Lee, Jiwoo Kim, Joo Yong Sim[†]
  Submitted to IEEE Robotics and Automation Letters (RA-L), under review
- **ModuLoop: Low-Level Code Generation Using Modular Synthesizer and Closed-Loop Debugger for Robotic Control**
  Gina Yoon[*], **Sumin Lee**[*], Joo Yong Sim[†]
  [*]Equal contribution (co-first authors)
  IEEE Robotics and Automation Letters (RA-L), vol. 10, no. 12, pp. 12812–12819, Dec. 2025.
  DOI: 10.1109/LRA.2025.3623437

Research Experience

2025.09 – Present **Graduate Researcher**, Autonomous Mechanical Systems Lab, Sookmyung Women's University (Advisor: Prof. Joo Yong Sim)

When to Code, When to Learn: Hybrid Code–Policy Generation
- LLM-planned programs combining code and RL policies per task segment
- Real-to-sim verification with teleport proxies before policy training
- 93.7% success on contact-rich table-leg assembly

Cube-in-Hand Multi-Camera Calibration (Ongoing)
- Robot-grasped AprilTag cube for joint multi-camera calibration
- 0.3 mm calibration error in held-out validation

2024.07 – 2025.08 **Undergraduate Research Intern**, Autonomous Mechanical Systems Lab, Sookmyung Women's University

ModuLoop: LLM-based Low-Level Robot Code Generation
- Built the simulation environment and LLM-based object recognition
- Conducted pick-and-place experiments

LLM-based Robotic Pipetting Automation
- Real-time coordinate extraction for precision liquid handling
- Automation using a 6-DoF manipulator

Patent Applications

- **Robot Control System and Method Based on Closed-Loop Modular Code Synthesis and Simulation Pre-Verification Using Large Language Models**
  대규모 언어 모델을 이용한 폐루프 모듈형 코드 합성 및 시뮬레이션 사전검증 기반 로봇 제어 시스템 및 방법
  KR Patent Application, filed 2025 (Co-inventor)
- **Automated 6-Axis Manipulator Pipette Precision Dispensing and Inspection System with LLM-based Data Processing and Real-Time Coordinate Extraction and Method Thereof**
  LLM 기반 데이터 처리 및 실시간 좌표 추출을 통한 6축 매니퓰레이터 피펫 정밀 분주 및 검토 자동화 시스템 및 그 방법
  KR Patent Application No. 10-2025-0024332, filed 2025 (Co-inventor)

Projects

- **Autonomous Indoor Mobile Robot**
  - ROS-based SLAM using LiDAR and a depth camera
  - Path planning and obstacle avoidance
- **Assistive Map for Socially Vulnerable Groups**
  - YOLO- and Detectron2-based perception
- **Smart Door System**
  - Raspberry Pi sensor integration and control

Work Experience

- **R&D Intern, MD Robot** 2024
  - Python-based robot UI development
  - Development of ROS packages and a BLE communication system

Awards

- Minister of Science and ICT Award, 4th X-Corps+ Festival, 2024

Skills

- **Programming:** Python, Kotlin (Android)

- **Robotics:** ROS, Isaac Sim, Isaac Lab
- **ML / Vision:** PyTorch, OpenCV
- **Development Tools:** Linux, Git, LaTeX